

Phase 1 (MVP) — Work Breakdown: phases → tasks → subtasks

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This document is the executable spine of the HelixVPN MVP. It decomposes the Phase-1 self-hostable MVP into stable, DB-ready work items (HVPN-P1-NNN), each carrying *description · depends-on · deliverable · acceptance · effort · required-test-types*, grouped by control-plane module + client core/UI + platform shims + deploy + QA. It is **spec-only** — it describes *what to build and how "done" is proven*, never the product itself. Authoritative companions: the data model / DDL / protobuf live in [02-control-plane.md](#) (cited here as [02]); the data plane in [01-data-plane.md](#) [01]; the client core + UI in [03-client-core-and-ui.md](#) [03]; security / privacy / PKI in [04-security-privacy-pki.md](#) [04]; product scope in [00-product-scope-and-principles.md](#) [00]. Primary research: [04_P1] (HelixVPN-Phase1-MVP.md), [04_P0] (HelixVPN-Phase0-Spike.md), [04_ARCH], [05_YBO], [SYNTHESIS].

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0. How to read this WBS

Granularity. Three levels: **Epic** (HVPN-P1-Enn, a workstream) → **Task** (HVPN-P1-NNN, a shippable unit, ~1–8 person-days) → **Subtask** (HVPN-P1-NNN.k, a single PR-sized change). Every **leaf** (a task with no subtasks, or a subtask) is a workable item per §11.4.93 and is the unit that lands in the SQLite single-source-of-truth DB (§3).

ID scheme (stable, append-only, §11.4.54). HVPN-P1- prefix; the numeric block encodes the epic: Enn → tasks nn0..nn9, e.g. E02 (store) → 020..029. Phase-0 entry gates use 001..006. IDs are **never renumbered**; new work appends at the next free number in the block (gaps allowed).

Field contract (every leaf).

Field	Meaning
Desc	What the item builds, in one or two sentences (§11.4.91 ≥6 words / ≥40 chars).
Deps	Hard upstream item IDs (must be complete first). – = none.
Deliverable	The concrete artefact(s) — file paths, binaries, generated clients.
Acceptance	The falsifiable PASS condition, captured-evidence per §11.4.5/.69/.107.
Effort	T-shirt + person-days: XS(1) S(2–3) M(4–6) L(7–10) XL(11–15).
Tests	

Field	Meaning
	Required test types from the §1 closed vocabulary (§11.4.169).

Risk ordering (§11.4.132). Within and across epics, the irreversible correctness floor (RLS, no-log lint, PKI revoke, kill-switch) is sequenced first; convenience UI last.

Anti-bluff (§11.4 / §11.4.27 / §11.4.107). Every Acceptance line is a *captured-evidence* assertion, not "code compiles". A leaf is complete only when its required-test-types are green **with** an evidence path recorded in the item's test_diary (§11.4.149) and, where it has a user-visible surface, a window-scoped MP4 recording vision-verified per §11.4.159/.163. Metadata-only / config-only / grep-without-runtime PASS is forbidden.

1. Required-test-types vocabulary (§11.4.169)

§11.4.169 mandates that **every** workable item declare the closed set of test types it must pass before closure; the universe of types is the §11.4.27 enumeration. The closed vocabulary + canonical abbreviations used in this WBS:

Abbrev	Type	Floor expectation for HelixVPN MVP
UNIT	Unit	pure logic (compiler, IPAM, delta diff, framing); mocks allowed only here (§11.4.27).
INT	Integration	real Postgres + Redis (testcontainers via containers submodule, §11.4.76); no mocks.
E2E	End-to-end	the netns rig [04_P0 §3] fed by the real control plane; curl reaches the LAN host.
FA	Full-automation	self-driving, re-runnable -count=3, no human in the loop (§11.4.98).
SEC	Security	RLS cross-tenant denial, key-never-leaves, mTLS, kill-switch, DNS-leak, secret-leak audit (§04).
CHAOS	Chaos	mid-flight SIGKILL / Redis drop / partial write; recovery restores consistent state (§11.4.85).
STRESS	Stress	sustained load + concurrency (≥10 parallel, ≥100 iters or ≥30 s) + boundary inputs (§11.4.85).

Abbrev	Type	Floor expectation for HelixVPN MVP
PERF	Performance	latency p50/p99 vs the SLO budget (§2); GC-tail watch on Go hot paths.
BENCH	Benchmarking	throughput / CPU-per-Gbps / handshakes-per-sec on the bench.sh rig [04_P0 §8].
SCALE	Scaling	N simulated agents holding streams; convergence + bounded memory over 24 h soak.
UI	UI	widget + golden tests on the three Flutter flavors.
UX	UX	flow walkthrough (enroll → connect → reach) with window-scoped MP4 + vision verdict (§11.4.159).
REC	Recorded-evidence	window-scoped MP4 (§11.4.154/.155) + media-validation pipeline verdict (§11.4.163).
CHAL	Challenge / HelixQA	a challenges / helix_qa bank entry scoring PASS only on captured evidence (§11.4.27/.107).

DDOS is **not-applicable** at MVP (single-node self-host, no public multi-tenant surface) — explicitly recorded NOT_APPLICABLE: single-node-selfhost per §11.4.6, re-armed in Phase 2 (HA / managed). Each task below lists only the **required** subset; absent types are out-of-scope for that item by design, not by omission.

2. MVP Definition-of-Done — the 8 acceptance gates + SLOs

The MVP ships only when **all eight** acceptance criteria (AC1–AC8) and **all four** SLOs (SLO1–SLO4) pass on a clean baseline (§11.4.40 full-suite retest, §11.4.108 runtime-signature on a fresh deploy). These are the *release gates* — restated verbatim from [04_P1 §11.2/§11.3] and bound to work-items in §21.



AC8 in [04_P1] bundles "no durable log" with "three apps drive it"; this WBS splits the app-coverage clause out as **AC9** for traceability — both are required.

AC1 — Self-host from zero. On a fresh rootless host, `helixvpncctl init + systemctl --user start helixvpn-pod` yields a running control plane + edge with generated tenant/admin/CA/overlay-/48/creds. *Evidence:* recorded terminal session, healthchecks green, `helix_up == 1`.

AC2 — Authorized reachability. Enroll one connector advertising a LAN + one client; the client's overlay namespace curls an authorized LAN host and gets the hello page. *Evidence:* E2E netns capture + pcap.

AC3 — Default-deny. The *same* client is denied a LAN host **not** granted by policy (connection refused/dropped at the edge verdict map). *Evidence:* negative E2E — packet dropped, no SYN-ACK.

AC4 — Transport auto-escalation. With plain WG UDP blocked (nft rule), the client escalates the TransportPolicy ladder to MASQUE/H3 and the tunnel stays up. *Evidence:* `StatusReport.transport == "masque-h3"`, tshark shows HTTP/3 (no WG signature), reachability preserved [04_P0 §5].

AC5 — Live policy reconfig. A Console/CLI policy edit reconfigures all affected devices in **< 1 s** with **zero process restarts**. *Evidence:* `helix_reconcile_seconds p99 < 1 s` histogram + Console live-event log.

AC6 — Fast revoke. Revoking a device removes its access in **< 1 s** (peer dropped from every map + kernel WG peer removed at edge + cert revoked). *Evidence:* `revoke→edge-enforcement` timing capture.

AC7 — Kill-switch + DNS-leak. On tunnel drop, **no plaintext egress and no DNS leak** occur (core state machine drives the firewall). *Evidence:* host-side pcap during forced drop shows zero leaked packets (§04, §11.4.107 liveness).

AC8 — No-logging as code. No durable connection/traffic/packet table exists; the CI schema-lint fails the build if one appears. *Evidence:* `schemalint green` + paired §1.1 mutation (connections table → FAIL) [02 §2.4].

AC9 — Three apps drive it. Access (iOS/Android/Linux), Connector daemon, and Console (web) each exercise AC1–AC7 through generated clients (no hand-written API code). *Evidence:* per-app UX recordings (§11.4.159).

SLOs (measured, alert-on-breach)

ID	SLO	Target	Item that wires the metric
SLO1	event → delta-on-wire	p99 < 1 s	HVPN-P1-073
SLO2		< 2 s	HVPN-P1-031

ID	SLO	Target	Item that wires the metric
	enroll → first NetworkMap		
SLO3	revoke → edge enforcement	< 1 s	HVPN-P1-033
SLO4	coordinator memory @ 10k streams	bounded, no leak / 24 h soak	HVPN-P1-074

3. Workable-item schema (§11.4.93 DB-ready)

Every leaf in §§5–20 maps 1:1 to a row in the git-tracked SQLite single source of truth (docs/.workable_items.db, §11.4.93/.95). Loader contract: a deterministic Go/SQL importer parses this Markdown's leaf blocks and upserts by atm_id (here HVPN-P1-NNN[.k]). The DDL the rows must satisfy:

```
-- docs/.workable_items.db – Phase-1 WBS projection (subset; full
  schema in 02/§11.4.93)
CREATE TABLE IF NOT EXISTS items (
  atm_id      TEXT PRIMARY KEY,           -- 'HVPN-
    P1-021'
  parent_id   TEXT REFERENCES items(atm_id), -- epic or
    parent task ('HVPN-P1-E02')
  kind        TEXT NOT NULL CHECK (kind IN
    ('epic', 'task', 'subtask')),
  title       TEXT NOT NULL CHECK (length(title) >= 6),
  description  TEXT NOT NULL CHECK (length(description) >= 40),
    -- §11.4.91/.148
  status      TEXT NOT NULL DEFAULT 'Queued'
    CHECK (status IN ('Queued', 'In progress', 'Ready
    for testing',
    'In
    testing', 'Reopened', 'Operator-blocked',
    'Obsolete (→
    Fixed.md)', 'Completed (→ Fixed.md)')),
  type        TEXT NOT NULL DEFAULT 'Task' CHECK (type IN
    ('Bug', 'Feature', 'Task')),
  severity    TEXT NOT NULL DEFAULT 'normal',
  epic        TEXT NOT NULL,             -- 'E02'
  module      TEXT NOT NULL,             --
    'store', 'coordinator', 'helix-core', ...
  deps        TEXT NOT NULL DEFAULT '[]', -- JSON array
    of atm_ids
  deliverable TEXT NOT NULL,
  acceptance  TEXT NOT NULL,
  effort_days INTEGER NOT NULL,
  test_types  TEXT NOT NULL,             -- JSON:
    ["UNIT", "INT", "SEC"] (§11.4.169)
```

```

dod_refs      TEXT NOT NULL DEFAULT '[]',          -- JSON:
               ["AC3","SL01"]
source_refs   TEXT NOT NULL DEFAULT '[]',          -- JSON:
               ["04_P1 §7","02 §7.2"]
created_at    TEXT NOT NULL DEFAULT (datetime('now')),
modified_at   TEXT NOT NULL DEFAULT (datetime('now'))
);
-- per-item test diary (§11.4.149) – PASS row REQUIRES an evidence
  path:
CREATE TABLE IF NOT EXISTS test_diary (
  id           INTEGER PRIMARY KEY AUTOINCREMENT,
  atm_id       TEXT NOT NULL REFERENCES items(atm_id),
  date_time    TEXT NOT NULL,
  tested_by    TEXT NOT NULL CHECK (tested_by IN
    ('User','Operator','AI-agent','HelixQA')),
  result       TEXT NOT NULL CHECK (result IN
    ('PASS','FAIL','SKIP')),
  test_type    TEXT NOT NULL,                    -- one of
    §11.4.169 vocab
  evidence_path TEXT,                            -- qa-results/
    <run-id>/...
  observations TEXT NOT NULL,
  CHECK (result <> 'PASS' OR (evidence_path IS NOT NULL AND
    length(evidence_path) > 0))
);

```

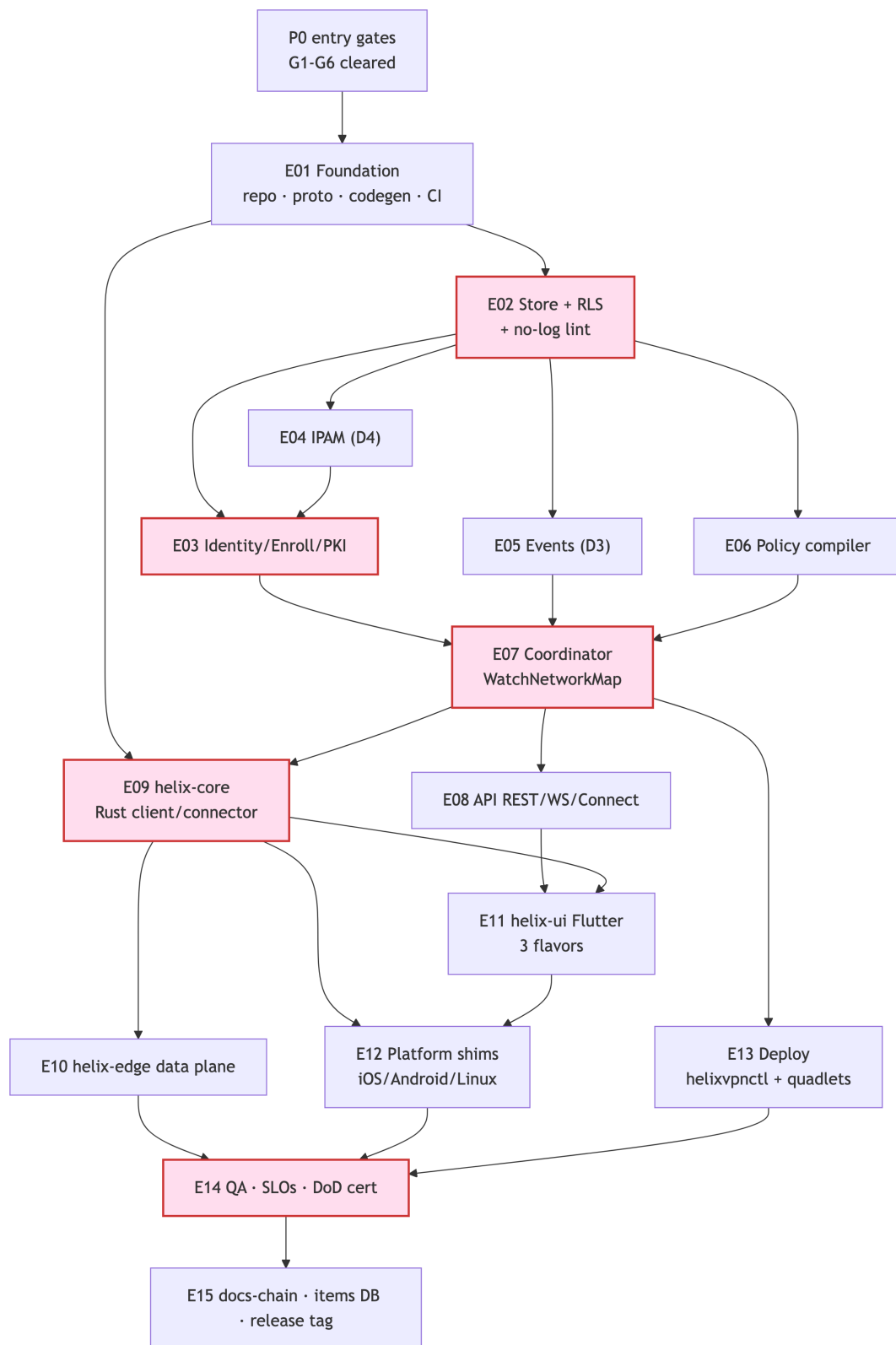
Example row (one leaf) so the projection is unambiguous:

```

INSERT INTO items
  (atm_id,parent_id,kind,title,description,epic,module,deps,
    deliverable,acceptance,effort_days,test_types,dod_refs,source_refs)
VALUES (
  'HVPN-P1-022','HVPN-P1-E02','task',
  'FORCE RLS + tenant_isolation on all 10 tenant tables',
  'Enable row-level security with a tenant_isolation USING/WITH
    CHECK policy on every tenant-scoped table and abort
    startup if the app role can bypass RLS.',
  'E02','store','["HVPN-P1-021"]',
  'migrations/0002_rls.sql; store/rls_guard.go startup check',
  'A query crafted to read tenant B from a tenant-A session
    returns ZERO rows (captured); startup aborts when role has
    rolsuper/rolbypassrls.',
  4,["UNIT","INT","SEC","CHAOS"],["AC8","AC2","AC3"],["02
    §2.3","04_P1 §2.2"]);

```

4. Epic map + dependency graph



Critical path (red): **E02** → **E03** → **E07** → **E09** → **E14**. The two Phase-0 make-or-break carries (iOS memory G3, edge language G4) are *resolved before Phase 1 starts* — Phase 1 assumes Rust core + the G4 edge decision are

settled [04_P0 §0]. Epics E04/E05/E06 fan out in parallel off E02 (disjoint scope → parallel PWUs, §11.4.58). E13 can start as soon as E07's container shape exists.

5. Phase 0 — entry gates (prerequisite)

Phase 0 is **not** Phase-1 work but is the entry condition; these items track the gate verdicts that unblock the MVP. If any gate fails, the linked Phase-1 epic re-plans (e.g. G3 fail → E12.iOS adopts the §6.4 fallback ladder).

HVPN-P1-E00 — Phase-0 spike gate certification epic · module: spike

- **HVPN-P1-001 — G1 plain-UDP data-path gate.** S(3) · deps: — · tests: BENCH, E2E
 - *Desc:* Certify Rust helix-core+boringtun moves real traffic client→gateway→connector LAN over plain UDP at ≥80% bare-link. *Deliverable:* bench.sh CSV + ping log [04_P0 §4/§8]. *Acceptance:* iperf3 through-tunnel ≥80% bare link; ping to LAN host succeeds. *DoD:* AC2 (precursor).
- **HVPN-P1-002 — G2 MASQUE-through-DPI gate.** M(5) · deps: HVPN-P1-001 · tests: BENCH, SEC, E2E
 - *Desc:* Certify the same core tunnels over MASQUE/H3 through a UDP block at ≥50% of plain, surviving 5% loss. *Deliverable:* tshark capture (classified HTTP/3, no WG signature) + goodput CSV. *Acceptance:* works with udp dport 51820 drop; ≥50% plain throughput; QUIC loss-recovery beats UoT strawman. *DoD:* AC4 (precursor).
- **HVPN-P1-003 — G3 iOS NE memory gate (make-or-break).** M(5) · deps: HVPN-P1-001 · tests: PERF, STRESS, REC
 - *Desc:* Certify the Rust core in NEPacketTunnelProvider stays under the device memory ceiling with ≥30% headroom over a 30-min / 1 GB transfer, both transports. *Deliverable:* Instruments RSS report (peak+steady) [04_P0 §6]. *Acceptance:* ≥30% headroom plain **and** MASQUE; else trigger §6.4 fallback decision (§11.4.66).
- **HVPN-P1-004 — G4 edge-language decision.** S(3) · deps: HVPN-P1-002 · tests: BENCH, PERF
 - *Desc:* Decide Rust vs Go for the MASQUE-terminating edge on benchmark numbers (CPU/Gbps, p99, churn, MASQUE effort). *Deliverable:* §12 decision-log row + A/B CSV. *Acceptance:* recorded decision with evidence; sets E10's language (default lean Rust if within ~10–15%) [04_P0 §7]. *Decision:* D5.
- **HVPN-P1-005 — G5 FFI boundary gate.** S(2) · deps: HVPN-P1-001 · tests: UI, E2E
 - *Desc:* Certify flutter_rust_bridge v2 drives connect/disconnect + live status from Dart. *Deliverable:* Flutter-Linux toggle recording.

Acceptance: status chip transitions

Connecting→Handshaking→Connected(masque,Nms) driven only by the core event stream. *Decision*: D2 (Rust core).

- **HVPN-P1-006 — G6 push-reconcile gate.** XS(1) · deps: HVPN-P1-001 · tests: E2E,FA
 - *Desc*: Certify the core converges to a static map delta with no restart (the seed of WatchNetworkMap). *Deliverable*: reconcile demo + map-delta log. *Acceptance*: editing map.json brings a new peer up with no process restart; reachability follows.
-

6. E01 — Foundation: repo, proto, codegen, CI floor

HVPN-P1-E01 — Foundation epic · module: repo

- **HVPN-P1-010 — Monorepo + submodule layout.** M(4) · deps: HVPN-P1-005 · tests: UNIT,FA *Desc*: Stand up the [SYNTHESIS §6] layout — helix-core/ helix-edge/ helix-go/ helix-proto/ helix-ui/ shims/ deploy/ — each decoupled, reusable, snake_case (§11.4.29), with upstreams/ + install_upstreams per repo (§11.4.36) and the incorporated submodules/ (containers, helix_qa, challenges, docs_chain, security, vision_engine). *Deliverable*: repo tree + .gitmodules + per-repo helix-deps.yaml (§11.4.31). *Acceptance*: install_upstreams configures every declared remote; a fresh clone builds each component skeleton; **no active CI** workflows present (§11.4.156, CM-N0-ACTIVE-CI green). *Tests*: FA bootstrap on a throwaway clone.
- **HVPN-P1-011 — helix-proto: agent protobuf + buf codegen.** M(4) · deps: HVPN-P1-010 · tests: UNIT,FA *Desc*: Author proto/ helix/agent/v1/agent.proto (the Coordinator service: Enroll, WatchNetworkMap, AdvertisePrefixes, ReportStatus, exactly as [02 §4] / [04_P1 §3]) and wire buf generate → Go + Dart + Rust stubs. *Deliverable*: generated gen/{go,dart,rust}/...; buf.gen.yaml; buf lint/buf breaking config. *Acceptance*: one source of truth → three generated clients; buf breaking gate fails on an incompatible change; no hand-written client code exists (§8 [04_P1]).
 - *Skeleton (the contract this WBS depends on — full text in [02 §4])*:

```
service Coordinator {  
  rpc Enroll(EnrollRequest) returns (EnrollResponse);  
  rpc WatchNetworkMap(WatchRequest) returns (stream  
    MapUpdate);  
  rpc AdvertisePrefixes(AdvertiseRequest) returns  
    (AdvertiseResponse);  
  rpc ReportStatus(StatusReport) returns (StatusAck);
```

```

}
// MapUpdate.oneof{ NetworkMap snapshot | MapDelta delta |
  KeepAlive } - peers PRE-FILTERED.

```

- **HVPN-P1-012 — helix-proto: REST OpenAPI + Dart/TS client gen.** S(3) · deps: HVPN-P1-011 · tests: UNIT, FA Desc: Define the OpenAPI 3.1 contract for the app-facing REST surface (/v1/enroll-tokens, /v1/devices, /v1/policies, /v1/networks, /v1/stream) and generate Dart + TS clients. *Deliverable*: openapi/helix.v1.yaml + generated clients. *Acceptance*: generated client round-trips a stub server; drift between spec and handler fails the gen check.
 - **HVPN-P1-013 — Local dev infra via containers submodule.** S(3) · deps: HVPN-P1-010 · tests: INT, FA Desc: On-demand Postgres+Redis for tests booted **only** through the containers submodule pkg/boot/pkg/compose/pkg/health (§11.4.76/.161, rootless Podman) — no ad-hoc docker. *Deliverable*: internal/testinfra wrapper; make test-infra-up/down. *Acceptance*: integration tests boot infra on demand; teardown leaves no orphan containers (§11.4.14).
-

7. E02 — Store, schema, RLS, no-log lint

HVPN-P1-E02 — Store + RLS + no-log lint (the truth floor) epic · module: store *Risk-first* (§11.4.132): *this epic is the irreversible correctness floor; it ships before any feature reads/writes data.*

- **HVPN-P1-020 — Migrations, enums, DB roles.** M(4) · deps: HVPN-P1-013 · tests: UNIT, INT Desc: Author the full DDL from [02 §2.2] / [04_P1 §2.1] via goose — tenants users devices connectors advertised_prefixes groups group_members policies overlay_pools device_certs audit_events + enums (user_role, device_kind); create three roles helix_owner (DDL), helix_app (non-super, non-bypass — runtime), helix_sys (system tasks). *Deliverable*: migrations/0001_core.sql.; sqlc.yaml. *Acceptance*: migrations apply+rollback clean on a fresh PG; \dt matches the spec; **no** connection/traffic table present.
- **HVPN-P1-021 — sqlc typed queries + store.WithTenant.** M(4) · deps: HVPN-P1-020 · tests: UNIT, INT Desc: Compile-time-checked queries via sqlc; WithTenant(ctx, tid, fn) runs each request inside a tx that SET LOCAL app.tenant_id (tx-scoped, can't leak across the pool) plus WithSystem for cross-tenant system tasks. *Deliverable*: internal/store/*.go, generated Queries. *Acceptance*: a query that forgets a WHERE tenant_id still returns only the active tenant's rows (proven by INT test).
- **HVPN-P1-022 — FORCE RLS + tenant_isolation policies.** M(4) · deps: HVPN-P1-021 · tests: UNIT, INT, SEC, CHAOS · DoD:

AC8,AC2,AC3 *Desc*: ENABLE+FORCE ROW LEVEL SECURITY and a tenant_isolation USING/WITH CHECK (tenant_id = current_setting('app.tenant_id')::uuid) on all 10 tenant tables; a startup guard aborts if helix_app has rolsuper/rolbypassrls.
Deliverable: migrations/0002_rls.sql; store/rls_guard.go.
Acceptance: SEC test — a crafted cross-tenant read returns **zero** rows (captured); startup aborts under a super role. CHAOS — connection drop mid-tx rolls back, no leak.

```
ALTER TABLE devices ENABLE ROW LEVEL SECURITY;
ALTER TABLE devices FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON devices
  USING (tenant_id = current_setting('app.tenant_id')::uuid)
  WITH CHECK (tenant_id =
    current_setting('app.tenant_id')::uuid);
-- repeat for users, connectors, advertised_prefixes, groups,
--   group_members,
--   policies, overlay_pools, device_certs,
--   audit_events
```

- **HVPN-P1-023 — No-logging CI schema-lint + mutation.** S(3) ·
deps: HVPN-P1-020 · tests: UNIT,SEC · DoD: AC8 *Desc*: tools/
schemalint fails the build if a durable table named/scoped like a
connection log appears (connections|flows|traffic|packets, or any
table carrying both a src/dst and a bytes/ts outside audit_events) [02
§2.4]. *Deliverable*: tools/schemalint/main.go + pre-build gate.
Acceptance: green on the real schema; **paired §1.1 mutation** — inject a
connections(src,dst,bytes,ts) table → lint FAILs (captured). This is
the mechanical enforcement of the privacy promise.
-

8. E03 — Identity, enrollment, PKI

HVPN-P1-E03 — Identity / Enrollment / PKI epic · module:
identity,pki *Risk-first*: revoke-<1 s and key-never-leaves are security-critical
(§04).

- **HVPN-P1-030 — Dual identity: OIDC + anonymous device tokens.**
M(5) · deps: HVPN-P1-022 · tests: UNIT,INT,SEC *Desc*: Managed
mode (tenant admins log in via any OIDC IdP — Keycloak/Authentik)
and anonymous mode (mint device enroll-tokens with no email/SSO —
the Mullvad "account number, no PII" posture) [04_P1 §6.1]. *Deliverable*:
internal/identity/; token mint/store-hashed/consume-atomic.
Acceptance: an anonymous device enrolls with zero PII persisted; an
OIDC user maps to a tenant/role; tokens are single-use and short-lived
(consume is atomic under concurrency — SEC).
- **HVPN-P1-031 — Enroll RPC (device key never leaves).**
M(5) · deps: HVPN-P1-030,HVPN-P1-040,HVPN-P1-032 · tests:

UNIT,INT,SEC,E2E · DoD: AC1,AC2; SLO2 Desc: Implement Coordinator.Enroll: verify token → IPAM-allocate overlay IP → issue short-lived device cert → persist device row (only the WG **public** key) → emit device.enrolled [04_P1 §6.2]. *Deliverable*: enroll handler + flow test. *Acceptance*: device generates its WG keypair locally; the control plane never receives a private key (SEC asserts absence in logs+DB); **enroll** → **first NetworkMap** < 2 s (SLO2, captured).

- **HVPN-P1-032 — PKI: device-cert issuance + tenant CA.** M(4) · deps: HVPN-P1-022 · tests: UNIT,INT,SEC Desc: internal/pki issues short-lived (24 h) mTLS device certs against a per-tenant CA; the CA root is the one true secret (KMS/offline backup) [04_P1 §6.3, §04]. *Deliverable*: pki.Issue/Rotate/Revoke; cert store. *Acceptance*: a cert authenticates the WatchNetworkMap channel; CA private key never appears in any app log (secret-leak audit, §11.4.10.A).
 - **HVPN-P1-033 — Cert rotation + revoke-<1 s.** M(4) · deps: HVPN-P1-032,HVPN-P1-070 · tests: UNIT,INT,SEC,PERF · DoD: AC6; SLO3 Desc: Auto-rotate before expiry over the existing channel; device.revoke drops the WG peer from every relevant map + enforces at the edge (kernel WG peer removed) + marks cert revoked, all within the convergence SLO. *Deliverable*: rotation scheduler + revoke path. *Acceptance*: **revoke** → **edge enforcement** < 1 s (SLO3, captured timing); a revoked cert is rejected immediately (SEC).
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9. E04 — IPAM (D4)

HVPN-P1-E04 — Overlay IPAM + subnet-collision (D4) epic · module: ipam

- **HVPN-P1-040 — ULA /48 provisioning + AllocOverlayIP.** M(4) · deps: HVPN-P1-022 · tests: UNIT,INT,STRESS Desc: On tenant create, provision an IPv6 ULA /48 (fd7a:helix:<rand>::/48, ::1 = gateway); AllocOverlayIP hands out the next host deterministically + concurrency-safe [02 §3.2]. **Decision D4: IPv6 ULA /48 per tenant** resolves RFC1918 collisions across N joined LANs (recommended over CGNAT-100.64 1:1 NAT) [SYNTHESIS §3 D4, 04_ARCH §3.4]. *Deliverable*: internal/ipam/. *Acceptance*: 1000 concurrent allocations yield no duplicate/gap (STRESS, captured); allocation is restart-stable (re-hydrates next_host).
- **HVPN-P1-041 — 4via6 for colliding IPv4 LANs.** M(5) · deps: HVPN-P1-040,HVPN-P1-060 · tests: UNIT,INT,E2E Desc: Map each connector site to a stable site-id and emit Tailscale-style 4via6 routes so a client may reach two connectors that both advertise 192.168.50.0/24 without collision. *Deliverable*: site-id allocator + 4via6 route synthesis feeding Peer.allowed_ips. *Acceptance*: E2E — one client reaches the *same* IPv4

CIDR served by two different connectors, disambiguated by site (captured curl to each).

10. E05 — Event backbone (D3)

HVPN-P1-E05 — Event backbone: Redis Streams (D3) epic · module: events

- **HVPN-P1-050 — events.Bus Redis Streams impl + envelope.** M(4) · deps: HVPN-P1-013 · tests: UNIT, INT, CHAOS *Desc:* Bus-agnostic events.Bus interface over Redis Streams: XADD produce, XReadGroup durable consumer groups, XACK, the §5.2 envelope (id, type, tenant_id, ts, actor, payload, trace_id). **Decision D3: Redis Streams for MVP**, NATS JetStream is the Phase-2 swap behind the same interface [SYNTHESIS §3 D3, 04_P1 §5]. *Deliverable:* internal/events/. *Acceptance:* at-least-once delivery with idempotent reactions; a replayed event is harmless (INT); a consumer crash mid-handle re-delivers on restart (CHAOS).
 - **HVPN-P1-051 — DLQ sweeper (XAUTOCLAIM) + metric.** S(3) · deps: HVPN-P1-050 · tests: UNIT, INT, CHAOS *Desc:* Reclaim stuck pending entries with XAUTOCLAIM, route exhausted-retry entries to a dead-letter, expose helix_events_dlq_total. *Deliverable:* sweeper goroutine + metric. *Acceptance:* a poisoned event lands in DLQ after N retries (captured), the metric increments, the live stream keeps flowing (CHAOS).
 - **HVPN-P1-052 — Event taxonomy producers.** S(3) · deps: HVPN-P1-050 · tests: UNIT, INT *Desc:* Wire the concrete taxonomy from [04_P1 §5.3] — device.enrolled/online/offline/revoked, connector.attached, connector.prefixes.changed, route.conflict.detected, policy.updated/compiled, gateway.failover — emitted by identity/registry/policy/edge. *Deliverable:* typed event constructors. *Acceptance:* each producer emits a well-formed envelope on its trigger (INT, captured stream read).
-

11. E06 — Policy model & compiler

HVPN-P1-E06 — Policy model + compiler epic · module: policy

- **HVPN-P1-060 — Spec parse + group/host resolution.** M(5) · deps: HVPN-P1-022 · tests: UNIT, INT, SEC *Desc:* Parse the Tailscale-ACL-flavored policies.spec jsonb (groups, hosts, acls, exitNodes) [04_P1 §7.1]; resolve group:* → device-id sets (via users + group_members) and host → CIDRs, cross-checking against

advertised_prefixes (emit route.conflict.detected on ambiguity).
Deliverable: internal/policy/parse.go, resolve.go. *Acceptance*: unknown group/host rejected; an ACL granting a revoked device rejected (SEC).

- **HVPN-P1-061 — Two-artifact compiler (AllowedIPs + verdict map).**

L(7) · deps: HVPN-P1-060 · tests: UNIT, SEC, PERF · DoD: AC2, AC3
Desc: Pure, deterministic, default-deny

compile(tenant, spec) → {visible, allowedIPs, exitNodes} [04_P1 §7.2]
producing **two** artefacts: coarse WG AllowedIPs (per-peer CIDR, data path) **and** a port-level verdict map (nftables/eBPF at the edge, since WG AllowedIPs is CIDR-only). Output is exactly what coordinator.buildMap consumes; peers are need-to-know filtered.

Deliverable: policy/compile.go + golden-table tests (spec → expected visible/allowedIPs). *Acceptance*: same spec ⇒ byte-identical output (determinism property test); a contractor granted only warehouse-cams:554,80 cannot reach port 22 (verdict-map SEC test); compile of a 1k-device tenant < 100 ms (PERF).

- **HVPN-P1-062 — Dry-run fail-closed validation + atomic activate/rollback.** M(4) · deps: HVPN-P1-061 · tests: UNIT, INT, SEC · DoD: AC5
Desc: policy.update runs the compiler in dry-run first; reject on unknown ref, a dst CIDR not covered by any advertised prefix, or a grant to a revoked device. Only a clean compile bumps policy.version, persists the compiled artefact, and emits policy.compiled; a prior version re-activates instantly (rollback) [04_P1 §7.3/§7.4]. *Deliverable*: validation + version activation. *Acceptance*: a bad policy never reaches the active version (fail-closed, captured); rollback restores the prior compiled set atomically (INT).

12. E07 — Coordinator & WatchNetworkMap

HVPN-P1-E07 — Coordinator + WatchNetworkMap epic · module: coordinator
Critical path; the < 1 s convergence promise lives here.

- **HVPN-P1-070 — In-memory per-tenant topology graph.** M(5) · deps: HVPN-P1-052, HVPN-P1-061, HVPN-P1-032 · tests: UNIT, INT, SCALE
Desc: Hydrate a per-tenant graph (devices, connectors, prefixes, groups, compiled policy) from Postgres on boot and keep it fresh by consuming events; the coordinator owns **no durable tables** [04_P1 §4.1]. *Deliverable*: internal/coordinator/graph.go. *Acceptance*: graph reconstructable from PG + events alone (proves Phase-2 stateless-coordinator seam); applying an event mutates only the affected subgraph (INT).
- **HVPN-P1-071 — buildMap(node) per-node map computation.** M(4) · deps: HVPN-P1-070 · tests: UNIT, SEC · DoD: AC2, AC3
Desc: buildMap returns NetworkMap{self, gateway, peers, dns, transport}

where peers = policy.PeersVisibleTo(node) (need-to-know) each with compiled AllowedIPs; relayEndpoint always routes via the gateway in MVP (hub-and-spoke, identical to the proven Phase-0 slice) [04_P1 §4.2]. *Deliverable*: coordinator/buildmap.go. *Acceptance*: a node never appears in another node's map unless policy grants it (SEC — the privacy invariant, captured).

- **HVPN-P1-072 — WatchNetworkMap stream loop.** L(7) · deps: HVPN-P1-071, HVPN-P1-080 · tests: UNIT, INT, E2E, CHAOS · DoD: AC5 *Desc*: Implement the streaming RPC: authenticate device by mTLS cert → send snapshot (or resume from known_version) → delta loop on the event subscription + keepalive ticker, with a **bounded** per-stream send queue (slow-consumer protection) [04_P1 §4.3]. *Deliverable*: coordinator/watch.go. *Acceptance*: reconnect with known_version=N resumes from deltas (no full resync, captured); a slow/stalled consumer is shed without blocking others (CHAOS).

```
func (c *Coordinator) WatchNetworkMap(ctx context.Context,
    req *connect.Request[agentv1.WatchRequest],
    stream *connect.ServerStream[agentv1.MapUpdate]) error {
    dev := authDevice(ctx) // mTLS device
    cert (HVPN-P1-032)
    sub := c.subscribe(dev.TenantID, dev.ID); defer sub.Close()
    _ = stream.Send(snapshotOrResume(c.buildMap(dev),
        req.Msg.KnownVersion))
    ka := time.NewTicker(20 * time.Second); defer ka.Stop()
    for {
        select {
        case <-ctx.Done(): return ctx.Err()
        case d := <-sub.C: if err :=
            stream.Send(deltaUpdate(d)); err != nil { return err }
        case <-ka.C: _ = stream.Send(keepAlive())
        }
    }
}
```

- **HVPN-P1-073 — Minimal-affected-set fan-out + reconcile metric.** M(5) · deps: HVPN-P1-072 · tests: UNIT, INT, PERF · DoD: AC5; SL01 *Desc*: On each event compute the **minimal** affected open-stream set (never broadcast a full resync on a small change) and enqueue per-stream deltas; instrument helix_reconcile_seconds from event-arrival to delta-on-wire [04_P1 §4.4]. *Deliverable*: fan-out + histogram. *Acceptance*: **event** → **delta-on-wire p99 < 1 s** (SL01, captured histogram); a 1-device policy change touches only the affected streams (INT count assertion).
- **HVPN-P1-074 — 24 h soak / 10k-stream memory bound.** M(5) · deps: HVPN-P1-073 · tests: SCALE, STRESS, PERF · DoD: SL04 *Desc*: Drive N=10k simulated agents holding streams while flapping policies; assert convergence SLO holds and coordinator RSS is bounded with no leak over 24 h. *Deliverable*: test/soak/ harness + RSS time-series.

Acceptance: SLO1 maintained under load; **coordinator memory bounded, no growth over 24 h** (SLO4, captured graph).

13. E08 — API surface + authz

HVPN-P1-E08 — API: Gin REST + Connect + WS/SSE + authz epic ·
module: api

- **HVPN-P1-080 — Connect handlers mounted on the Gin listener + mTLS resolver.** M(4) · deps: HVPN-P1-011, HVPN-P1-032 · tests: UNIT, INT, SEC *Desc:* One multiplexed server: Connect handlers (HTTP/2 native agents, downgrade for browsers) mounted alongside Gin REST; `authDevice(ctx)` resolves the device identity from its mTLS cert [04_P1 §8]. *Deliverable:* `internal/api/server.go`, `authz_device.go`. *Acceptance:* a native agent and a gRPC-Web caller both reach Coordinator; an unauthenticated agent RPC is rejected (SEC).
 - **HVPN-P1-081 — REST routes + OpenAPI + RBAC.** M(5) · deps: HVPN-P1-080, HVPN-P1-012 · tests: UNIT, INT, SEC *Desc:* Gin routes (`/v1/enroll-tokens`, `/v1/devices`, `/v1/policies`, `/v1/networks`, `/v1/audit`) behind OIDC/session-or-API-token + RBAC (admin/operator/member), with RLS as the backstop. *Deliverable:* route handlers + RBAC middleware. *Acceptance:* a member token cannot mint enroll-tokens (403, SEC); RLS prevents cross-tenant reads even if RBAC is bypassed (defense-in-depth INT).
 - **HVPN-P1-082 — GET `/v1/stream` live WS/SSE fan-out.** S(3) · deps: HVPN-P1-081, HVPN-P1-052 · tests: UNIT, INT, E2E *Desc:* Stream control-plane events (`device.online`, `route.changed`, `handshake.failing`, `policy v N active`) to the Console over WebSocket/SSE [04_P1 §8]. *Deliverable:* `api/stream.go`. *Acceptance:* a policy apply surfaces a live "policy v N active; K devices updated" event to a connected Console within the convergence window (E2E, captured).
 - **HVPN-P1-083 — AdvertisePrefixes + ReportStatus RPCs.** S(3) · deps: HVPN-P1-080, HVPN-P1-052 · tests: UNIT, INT, SEC *Desc:* `AdvertisePrefixes` accepts connector CIDRs (also settable via Console) emitting `connector.prefixes.changed`; `ReportStatus` carries presence + current transport + rtt **only** (deliberately no bytes/flows/destinations — the no-logging invariant on the wire). *Deliverable:* both handlers. *Acceptance:* a status report persists no traffic data (SEC asserts the message has no byte/flow fields); advertised prefixes flow into the coordinator graph (INT).
-

14. E09 — helix-core (Rust client/connector)

HVPN-P1-E09 — helix-core: Rust client/connector epic · module:
helix-core *The shared client/connector/edge crate; the durable evolution of the Phase-0 interfaces.*

- **HVPN-P1-090 — Transport trait + plain-UDP + MASQUE impls.** L(8) · deps: HVPN-P1-002 · tests: UNIT, BENCH, SEC, E2E · DoD: AC4
Desc: Harden the Phase-0 Transport trait (send/recv/kind/effective_mtu, unreliable datagrams) + plain_udp + masque (quinn + h3 + hand-rolled CONNECT-UDP / HTTP-Datagram framing per RFC 9298/9297/9221) into the production crate helix-transport [04_P0 §4.2/§5, 03 §transport]. **Decision D1: MASQUE/QUIC primary obfuscating transport** (true Mullvad parity, single Rust impl) — surfaced against the Hysteria2 alternative [SYNTHESIS §3 D1].
Deliverable: crates/helix-transport/. *Acceptance:* MASQUE flow classifies as HTTP/3 with no WG signature (tshark, SEC); ≥50% plain throughput (BENCH); same trait drives all three consumers byte-for-byte.
- **HVPN-P1-091 — helix-wg boringtun wrapper.** M(5) · deps: HVPN-P1-090 · tests: UNIT, E2E, STRESS *Desc:* Production wrapper around boringtun Tunn (handshake, encrypt/decrypt, timer tick, the four WgVerdict outputs) — pure-userspace WG identical on Linux/Android/iOS [04_P0 §4.4]. *Deliverable:* crates/helix-wg/. *Acceptance:* sustained 1 GB transfer with no handshake stall (STRESS); rekey at the 2-min boundary works (captured).
- **HVPN-P1-092 — Orchestrator + network-map reconciler.** L(7) · deps: HVPN-P1-091, HVPN-P1-072 · tests: UNIT, E2E, FA, CHAOS · DoD: AC5 *Desc:* The three-loop orchestrator (tun→wg→transport, transport→wg→tun, timer) + a reconciler that consumes streamed MapUpdate (snapshot/delta) and converges peers/AllowedIPs/transport **without restart**; client vs connector routing posture via mode [04_P0 §4.5, 04_P1 §3]. *Deliverable:* crates/helix-core/. *Acceptance:* a streamed delta adds/removes a peer with no reconnect of unrelated state (FA); a mid-stream coordinator restart re-syncs from known_version (CHAOS).
- **HVPN-P1-093 — Auto-escalation ladder.** M(5) · deps: HVPN-P1-090, HVPN-P1-092 · tests: UNIT, E2E, SEC · DoD: AC4 *Desc:* Walk TransportPolicy.order (["plain-udp", "lwo", "masque-h3"]) on handshake failure, honoring allow_user_override; emit transport in ReportStatus. *Deliverable:* ladder state machine. *Acceptance:* with plain WG blocked the client escalates to MASQUE and stays up (E2E, captured transport=="masque-h3"); ladder is deterministic across N runs (§11.4.50).
- **HVPN-P1-094 — Kill-switch + DNS-leak protection.** L(7) · deps: HVPN-P1-092 · tests: UNIT, E2E, SEC, CHAOS · DoD: AC7 *Desc:* A core

state machine drives the platform firewall so that on tunnel Down/
Reconnecting **no plaintext egress and no DNS leak** occur; DNS is
pinned to the overlay resolver while up [04_P1 §11.2, §04]. *Deliverable:*
kill-switch module + per-OS firewall hooks (called by shims). *Acceptance:*
host-side pcap during a forced drop shows **zero** leaked packets and zero
plaintext DNS (SEC, captured); state survives an abrupt interface flap
(CHAOS).

- **HVPN-P1-095 — helix-ffi (flutter_rust_bridge v2) surface.** M(4) ·
deps: HVPN-P1-092 · tests: UNIT, UI, FA *Desc:* The tiny stable FFI:
start(ClientConfig), stop(),
status_stream(StreamSink<TunnelStatus>) mirroring the core
broadcast channel (Connecting|Handshaking|
Connected{transport, rtt}|Reconnecting|Down{reason}) [04_P0 §9].
Deliverable: crates/helix-ffi/ + generated Dart bindings. *Acceptance:*
Dart drives connect/disconnect and renders live status from the stream
alone (UI golden + FA re-run ×3).
-

15. E10 — helix-edge (data plane)

HVPN-P1-E10 — helix-edge data plane epic · module: helix-edge
*Language fixed by Phase-0 G4 (HVPN-P1-004); default lean Rust sharing heli
x-transport [04_P0 §7].*

- **HVPN-P1-100 — MASQUE termination on :443/udp + masquerade.**
L(8) · deps: HVPN-P1-090, HVPN-P1-004 · tests:
UNIT, BENCH, SEC, E2E · DoD: AC4 *Desc:* Terminate MASQUE/H3, extract
HTTP Datagrams → WG → kernel WG; serve a believable decoy site to
anything that isn't a valid CONNECT-UDP flow (native camouflage)
[04_P0 §5.4, 01]. *Deliverable:* helix-edge/ listener. *Acceptance:* a scanner
hitting :443 sees a plain website; a valid client tunnels through (SEC +
E2E, captured); CPU/Gbps within the G4 budget (BENCH).
- **HVPN-P1-101 — Kernel-WG peer programming + relay
(hub-and-spoke).** M(5) · deps: HVPN-P1-100, HVPN-P1-072 · tests:
UNIT, INT, E2E · DoD: AC2 *Desc:* Program kernel WG peers from the
coordinator's map and relay peer[?]peer traffic via the gateway (MVP
hub-and-spoke; direct P2P is Phase 2) [04_P1 §4.2]. *Deliverable:*
edge[?]coordinator sync + WG config apply. *Acceptance:* client reaches
an authorized connector LAN host through the relay (E2E, captured
curl).
- **HVPN-P1-102 — Edge verdict-map enforcement (nftables/eBPF) +
revoke drop.**
M(5) · deps: HVPN-P1-061, HVPN-P1-101, HVPN-P1-033 · tests:
UNIT, INT, SEC, PERF · DoD: AC3, AC6; SL03 *Desc:* Apply the policy
compiler's port-level verdict map at the edge (the port granularity WG
AllowedIPs lacks) and drop a revoked device's sessions on

device.revoked [04_P1 §7.2]. *Deliverable*: nftables/eBPF programmer.
Acceptance: an unauthorized port is dropped (AC3 SEC, captured);
revoke removes the kernel peer < 1 s (SLO3, captured); enforcement
adds negligible p99 (PERF).

- **HVPN-P1-103 — Edge hardening posture.**

S(3) · deps: HVPN-P1-100 · tests: SEC, CHAOS *Desc*: Rootless
Podman, read-only rootfs, seccomp, NET_ADMIN-only, **no SSH** in the edge
container, fail-static (Go/control never in the packet path) [SYNTHESIS
§7, §04]. *Deliverable*: edge container manifest + seccomp profile.
Acceptance: the container runs unprivileged except NET_ADMIN; killing
the control plane does **not** drop established data-plane tunnels
(fail-static CHAOS, captured).

16. E11 — helix-ui (Flutter) + design system

HVPN-P1-E11 — helix-ui: Flutter design system + 3 flavors epic ·
module: helix-ui *All UI under OpenDesign tokens (§11.4.162) — light+dark,*
no ad-hoc CSS; Riverpod state = pure function of the core status stream [03,
04_ARCH §5].

- **HVPN-P1-110 — helix_design system (OpenDesign tokens, Material**

3). M(5) · deps: HVPN-P1-010 · tests: UNIT, UI, REC *Desc*:
Brand-tokenized design system (connection-state palette, signature
components: ConnectButton, StatusChip, ExitPicker,
ShieldIndicator, AdaptiveScaffold) with light+dark variants via
OpenDesign (§11.4.162), no overlapping labels. *Deliverable*: helix-ui/
packages/helix_design/. *Acceptance*: visual-regression goldens pass for
every component in both themes (UI); a window-scoped MP4 shows the
palette reacting to connection state (REC, vision-verified §11.4.159).

- **HVPN-P1-111 — Shared shell runHelixApp(flavor, ...) + Riverpod**
core-status binding. M(4) · deps: HVPN-P1-110, HVPN-P1-095 ·
tests: UNIT, UI, FA *Desc*: One tree → three flavors via
runHelixApp(flavor, home, capabilities) (Access / Connector /
Console), with Riverpod providers derived purely from the helix-ffi
status stream [03]. *Deliverable*: helix-ui/app/. *Acceptance*: the same
widget tree renders all three flavors; UI state is a deterministic function
of a replayed status stream (FA).

- **HVPN-P1-112 — Access app screens (enroll → connect → reach).**

L(7) · deps: HVPN-P1-111, HVPN-P1-012 · tests: UI, UX, E2E, REC ·
DoD: AC9 *Desc*: End-user flow: paste/scan enroll token → connect →
ExitPicker / authorized-network list → live status. *Deliverable*: Access
flavor screens. *Acceptance*: a real enroll→connect→reach-LAN journey
recorded as a window-scoped MP4 with a vision verdict that the
authorized host loaded (UX/REC, §11.4.159/.143); golden tests for each
screen (UI).

- **HVPN-P1-113 — Console (web) admin screens.** L(7) · deps: HVPN-P1-111, HVPN-P1-082 · tests: UI, UX, E2E, SEC, REC · DoD: AC5, AC9
Desc: The Web-only flavor (no core_ffl): devices, enroll-tokens, policy editor (validate→activate→rollback), live event feed via /v1/stream.
Deliverable: Console flavor. *Acceptance:* a policy edit in the Console reflects on devices < 1 s with a live "policy v N active" feed (E2E/REC, AC5); RBAC hides admin actions from member (SEC).
 - **HVPN-P1-114 — Connector daemon UI/headless control.** M(4) · deps: HVPN-P1-111, HVPN-P1-083 · tests: UI, E2E, REC · DoD: AC9
Desc: The Connector flavor: enroll, advertise CIDRs, show advertised prefixes + reachability/health (runs headless on a host with a thin status UI). *Deliverable:* Connector flavor + daemon mode. *Acceptance:* a connector advertises a LAN and its prefixes appear in the Console + a client's map (E2E, captured).
-

17. E12 — Platform tunnel shims (iOS/Android/Linux)

HVPN-P1-E12 — Platform TunnelPlatform shims epic · module: shims
*MVP platforms only: iOS, Android, Linux (Windows/macOS = Phase 2; HarmonyOS/Aurora = Phase 3). The shim is the **only** platform-specific code [SYNTHESIS §5].*

- **HVPN-P1-120 — Linux tun shim (Rust-native).** S(3) · deps: HVPN-P1-092 · tests: E2E, FA · DoD: AC1, AC2
Desc: Linux core drives the TUN directly (kernel WG or userspace); the reference platform for the netns E2E rig. *Deliverable:* shims/linux/. *Acceptance:* the Access Linux build runs the full enroll→connect→reach journey on the netns rig (E2E).
- **HVPN-P1-121 — Android VpnService + JNI shim.** M(5) · deps: HVPN-P1-092, HVPN-P1-094 · tests: E2E, UX, REC, SEC · DoD: AC7, AC9
Desc: VpnService owns the TUN; JNI loads helix-core.so; the kill-switch maps to VpnService always-on/lockdown semantics. *Deliverable:* shims/android/ (Kotlin+JNI). *Acceptance:* tunnel up on a real phone (REC device recording); kill-switch blocks plaintext on drop (SEC, captured).
- **HVPN-P1-122 — iOS NEPacketTunnelProvider shim (within G3 budget).** L(8) · deps: HVPN-P1-003, HVPN-P1-092, HVPN-P1-094 · tests: E2E, UX, REC, PERF, SEC · DoD: AC7, AC9
Desc: NEPacketTunnelProvider (Swift) configures NEPacketTunnelNetworkSettings and pumps packetFlow [?] the Rust core; honor the G3 memory ceiling (apply §6.4 fallbacks only if G3 demanded them) [04_P0 §6]. *Deliverable:* shims/apple/ios/. *Acceptance:* tunnel up on a real device within the memory ceiling over a sustained

transfer (PERF/REC, captured Instruments RSS); kill-switch + DNS-leak hold (SEC).

18. E13 — Deploy: helixvpncctl + Podman quadlets + containers submodule

HVPN-P1-E13 — Deploy & operations epic · module: deploy

- **HVPN-P1-130 — helixvpncctl bootstrap + ops CLI (Cobra).** M(5) ·
deps: HVPN-P1-020, HVPN-P1-130_dep · tests: UNIT, FA, E2E · DoD:
AC1 Desc: The ops CLI replacing bash installers: init (generates tenant, admin, CA root, overlay /48, PG/Redis creds, gateway WG keys, quadlets), gateway keys, enroll-token, policy apply, device revoke [04_P1 §9.1]. *Deliverable:* cmd/helixvpncctl/. *Acceptance:* helixvpncctl init on a fresh host produces a runnable deploy (FA, captured); policy apply ./policy.jsonc validates + activates (E2E).

```
helixvpncctl init --domain gw.example.com --data-dir /var/lib/helixvpn
helixvpncctl gateway keys
helixvpncctl enroll-token --kind connector --site warehouse #
prints token / QR
helixvpncctl policy apply ./policy.jsonc
# dry-run compile → activate
helixvpncctl device revoke <id>
```

- **HVPN-P1-131 — Podman quadlets (rootless, systemd-managed).** M(4) · deps: HVPN-P1-130, HVPN-P1-103 · tests: INT, FA, SEC · DoD:
AC1 Desc: Render the pod quadlets (helixd, helix-edge, postgres, redis) — rootless, read-only rootfs, helixd connects to PG as the **non-superuser** role so RLS is enforced [04_P1 §9.2, §11.4.161]. *Deliverable:* deploy/quadlet/*.container, helixvpn.pod. *Acceptance:* systemctl --user start helixvpn-pod brings the stack up rootless (FA); the app role cannot bypass RLS (SEC).

```
# deploy/quadlet/helixd.container (rootless; read-only rootfs)
[Container]
Image=ghcr.io/helixdevelopment/helixd:1.0
Pod=helixvpn.pod
Environment=DATABASE_URL=postgres://helix_app@localhost/helix
# NON-superuser → RLS enforced
Environment=REDIS_URL=redis://localhost:6379
ReadOnly=true
[Container] # helix-edge — winner of Phase-0 G4
Image=ghcr.io/helixdevelopment/helix-edge:1.0
Pod=helixvpn.pod
AddCapability=NET_ADMIN # :443/udp MASQUE + kernel WG; no
SSH in this container
```

- **HVPN-P1-132 — On-demand integration infra via containers submodule.** S(3) · deps: HVPN-P1-013, HVPN-P1-131 · tests: INT, FA *Desc:* Wire the §14 QA harness to boot the full stack on demand through the containers submodule (§11.4.76) so integration/E2E never require a manual podman up. *Deliverable:* deploy/integration/. *Acceptance:* the E2E suite boots+tears down the stack itself (FA, captured).
 - **HVPN-P1-133 — Compose dev-loop + K8s Phase-2 shape (reference).** S(2) · deps: HVPN-P1-131 · tests: INT *Desc:* A Docker Compose dev-loop equivalent (local only) and a stateless-coordinator Kubernetes manifest **as reference** proving the Phase-1 seams permit HA without reshaping [02 §11.2/§11.3]. *Deliverable:* deploy/compose/dev.yml, deploy/k8s/helixd-deployment.yaml. *Acceptance:* dev compose boots PG+Redis+helixd locally; the K8s manifest is lint-valid (not deployed at MVP).
-

19. E14 — QA, SLOs, DoD certification

HVPN-P1-E14 — QA · SLOs · DoD certification epic · module: qa
Anti-bluff capstone: this epic owns the §11.4.40 full-suite retest, the §11.4.107 recorded evidence, and the §11.4.27 100%-type coverage that turn the 8 ACs from claims into proofs.

- **HVPN-P1-140 — netns + DPI + netem E2E rig (CI-ready).** M(5) · deps: HVPN-P1-101, HVPN-P1-120 · tests: E2E, FA *Desc:* Promote the Phase-0 rig to a reproducible, fully-automated harness now fed by the **real** control plane: lanA netns service, nft WG-block for AC4, tc netem loss/delay, make e2e [04_P0 §3]. *Deliverable:* test/e2e/rig/. *Acceptance:* make e2e runs the full slice unattended and re-runs -count=3 identically (FA, §11.4.98).
- **HVPN-P1-141 — SLO instrumentation + dashboards.** S(3) · deps: HVPN-P1-073, HVPN-P1-074, HVPN-P1-031, HVPN-P1-033 · tests: PERF, FA · DoD: SL01-4 *Desc:* Prometheus metrics (helix_reconcile_seconds, enroll/revoke timers, coordinator RSS) + Grafana-as-code dashboards + alert rules at the §2 SLO targets. *Deliverable:* deploy/observability/. *Acceptance:* all four SLOs measured and asserted in CI thresholds (captured histograms, alert-on-breach).
- **HVPN-P1-142 — Security test suite (RLS, key-never-leaves, mTLS, kill-switch, leak audit).** L(7) · deps: HVPN-P1-022, HVPN-P1-031, HVPN-P1-094, HVPN-P1-102 · tests: SEC, CHAOS, SCALE · DoD: AC3, AC6, AC7, AC8 *Desc:* Consolidate the security-critical assertions into one suite driven by the security submodule tooling (§04): cross-tenant RLS denial, private-key absence, mTLS rejection of bad certs, kill-switch/DNS-leak pcap, revoke timing, secret-leak repo audit (§11.4.10.A). *Deliverable:* test/security/. *Acceptance:* every assertion

green with captured evidence; paired §1.1 mutations (drop an RLS policy / weaken kill-switch) FAIL their gate.

- **HVPN-P1-143 — Stress + chaos suite (§11.4.85).** M(5) · deps: HVPN-P1-072, HVPN-P1-050, HVPN-P1-092 · tests: STRESS, CHAOS, SCALE *Desc:* Sustained-load + concurrency + boundary + failure-injection across the control plane (Redis kill mid-event, PG drop mid-tx, coordinator SIGKILL with open streams, disk-full) with categorized recovery [§11.4.85]. *Deliverable:* test/stress_chaos/. *Acceptance:* every fault recovers to a consistent state with a captured recovery_trace.log; no deadlock/leak under 10× concurrency.
 - **HVPN-P1-144 — HelixQA Challenge bank + vision-verified recordings.** M(5) · deps: HVPN-P1-112, HVPN-P1-113, HVPN-P1-140 · tests: CHAL, REC, UX · DoD: AC1-AC9 *Desc:* Author a challenges/helix_qa bank where each Challenge dispatches a real user journey and scores PASS only on captured evidence; every user-visible AC gets a window-scoped MP4 run through the §11.4.163 media-validation pipeline (vision verdict vs SPECIFY-phase patterns) [§11.4.27/.107/.158/.159/.160]. *Deliverable:* submodules/challenges/banks/helixvpn.yaml + docs/qa/<run-id>/. *Acceptance:* the 8-AC journey set scores PASS with a vision-confirmed MP4 per AC (no frozen/bluff frame, §11.4.107); a deliberately broken build scores FAIL (analyzer self-validated golden-good/bad).
 - **HVPN-P1-145 — Full-suite retest + DoD certification gate.** M(4) · deps: HVPN-P1-140, HVPN-P1-141, HVPN-P1-142, HVPN-P1-143, HVPN-P1-144 · tests: FA, E2E, CHAL · DoD: AC1-AC9, SL01-4 *Desc:* The §11.4.40 release-gate sweep on a clean baseline (pre-build + post-build + on-device + meta-mutation + Challenge bank + no-log lint + CONTINUATION sync), asserting all 8 ACs + 4 SLOs end-to-end on a fresh §11.4.108 deploy. *Deliverable:* scripts/release_gate.sh + the certification report under docs/qa/release/. *Acceptance:* all of §21 green simultaneously on a from-zero install; any red blocks the tag.
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20. E15 — Docs-chain, workable-items DB, release

HVPN-P1-E15 — Docs-chain, items DB, release epic · module: governance

- **HVPN-P1-150 — Workable-items DB projection + docs-chain sync.** M(4) · deps: HVPN-P1-010 · tests: UNIT, INT, FA *Desc:* Implement the §3 loader: parse this WBS's leaves → upsert items/test_diary in docs/.workable_items.db (git-tracked, §11.4.95) and register the docs-chain context so HTML/PDF/DOCX exports + the DB stay in lockstep (§11.4.106/.93/.65/.153). *Deliverable:* cmd/workable-items/ +

.docs_chain/contexts/wbs.yaml. *Acceptance*: md5db byte-identical round-trip (§11.4.93); a leaf edit re-syncs exports out-of-the-box; verify is the CI gate.

- **HVPN-P1-151 — Release tagging + multi-upstream publish.** S(3) · deps: HVPN-P1-145, HVPN-P1-150 · tests: FA · DoD: AC1-AC9 *Desc*: Cut the prefixed release tag (<HELIX_RELEASE_PREFIX>-1.0.0-mvp from .env else snake_case root dir, §11.4.151) on the main repo **and** every owned submodule with the identical prefix, published to all upstreams via merge-onto-latest-main (no force-push, §11.4.113). *Deliverable*: the tag set + changelog under docs/changelogs/. *Acceptance*: git tag -l '<PREFIX>-*' enumerates the whole release across repos; the DoD certification report (HVPN-P1-145) is the tag's evidence; the tag is reachable on every remote.

21. DoD [?] work-item traceability matrix

Each release gate must trace to at least one *implementing* item and one *proving* (test/QA) item. A gate with no green proving item cannot ship (§11.4.123 rock-solid proof).

Gate	Implementing items	Proving items
AC1 self-host from zero	HVPN-P1-130, 131, 132	HVPN-P1-144, 145
AC2 authorized reach	HVPN-P1-031, 071, 101, 041	HVPN-P1-140, 144
AC3 default-deny	HVPN-P1-061, 071, 102	HVPN-P1-142, 144
AC4 transport auto-escalate	HVPN-P1-090, 093, 100	HVPN-P1-002, 140, 144
AC5 policy edit < 1 s	HVPN-P1-062, 072, 073, 113	HVPN-P1-141, 144
AC6 revoke < 1 s	HVPN-P1-033, 102	HVPN-P1-141, 142
AC7 kill-switch + DNS-leak	HVPN-P1-094, 121, 122	HVPN-P1-142, 144
AC8 no durable log	HVPN-P1-020, 023, 083	HVPN-P1-023 (mutation), 142
AC9 three apps drive it	HVPN-P1-112, 113, 114	HVPN-P1-144
SLO1 event→delta p99<1s	HVPN-P1-073	HVPN-P1-141
SLO2 enroll→map <2s	HVPN-P1-031	HVPN-P1-141

Gate	Implementing items	Proving items
SLO3 revoke→edge <1s	HVPN-P1-033, 102	HVPN-P1-141, 142
SLO4 coord mem @10k	HVPN-P1-074	HVPN-P1-141, 143

22. Effort roll-up & critical path

Indicative person-day totals (mid-point of the T-shirt range) for sizing, **not** a commitment — Phase-0 carries (G3/G4) are excluded as prerequisites.

Epic	Items	~Person-days
E00 Phase-0 gates (prereq)	6	19
E01 Foundation	4	14
E02 Store + RLS + lint	4	15
E03 Identity/Enroll/PKI	4	18
E04 IPAM	2	9
E05 Events	3	10
E06 Policy compiler	3	16
E07 Coordinator	5	26
E08 API + authz	4	15
E09 helix-core (Rust)	6	36
E10 helix-edge	4	21
E11 helix-ui (Flutter)	5	27
E12 Platform shims	3	16
E13 Deploy	4	14
E14 QA · SLOs · DoD	6	29
E15 Governance/release	2	7
MVP total (excl. E00)	59	~273

Critical path (longest dependency chain to AC certification):

E01(010→011) → E02(020→021→022) → E03(032) → E07(070→071→072→073) → E09(092→093/094) → E10(100→101→102) → E14(140→142→144→145) → E15(151). The two parallelizable fan-outs that keep wall-clock down: **E04/E05/E06** off E02 (disjoint scope → concurrent PWUs §11.4.58), and **E11 UI** off E08/E09 in parallel with E10/E12. Audio-style "top priority first" maps here to the **security/correctness floor** (E02 RLS, E03 revoke, E09 kill-switch) leading every iteration per §11.4.132.

Open decisions surfaced (not silently resolved), each with a recommendation and the gate that finalizes it: **D1** MASQUE/QUIC primary (rec: yes — Mullvad parity, HVPN-P1-090) · **D2** Rust core (rec: yes — G3/G5, settled) · **D3** Redis Streams MVP → NATS Phase 2 (rec: yes, HVPN-P1-050) · **D4** IPv6 ULA /48

- 4via6 (rec: yes — HVPN-P1-040/041) · **D5** edge language (rec: Rust if within ~10–15%, decided by G4/HVPN-P1-004) · **D6** asymmetric per-leg transport (rec: defer to Phase 2; MVP is single-transport hub-and-spoke) · **D7** MVP ambition (rec: lean tunnel-first MVP, this WBS) [SYNTHESIS §3].
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Sources

[04_P1] HelixVPN-Phase1-MVP.md — control-plane modular monolith, DDL+RLS, the Coordinator protobuf + WatchNetworkMap, Redis-Streams event taxonomy, policy compiler, enrollment/PKI, helixvpnctl + quadlets, the §11.2 8-criterion DoD + §11.3 SLOs + §11.4 no-log lint (the spine of this WBS). · [04_P0] HelixVPN-Phase0-Spike.md — G1–G6 entry gates, the Transport trait, boringtun wrapper, MASQUE/CONNECT-UDP, iOS memory harness, Go-vs-Rust edge, flutter_rust_bridge FFI, netns+DPI+netem rig. · [04_ARCH] HelixVPN-Architecture-Refined.md — control/data separation, ULA /48 + 4via6 (D4), no-logging, default-deny, repo layout. · [02] 02-control-plane.md — authoritative DDL, RLS, IPAM, protobuf, events, coordinator, API, deploy manifests (this WBS references, never re-defines). · [01] 01-data-plane.md, [03] 03-client-core-and-ui.md, [04] 04-security-privacy-pki.md, [00] 00-product-scope-and-principles.md. · [05_YBO] Go+Gin+PG+Redis+Podman mandate. · [SYNTHESIS] §§1–9 — stack floor, the D1–D7 key decisions, ecosystem wiring (containers/helix_qa/challenges/docs_chain/security/vision_engine), constitution bindings (§11.4.27/.40/.58/.76/.85/.93/.106/.107/.113/.132/.151/.156/.159/.161/.162/.163/.169).